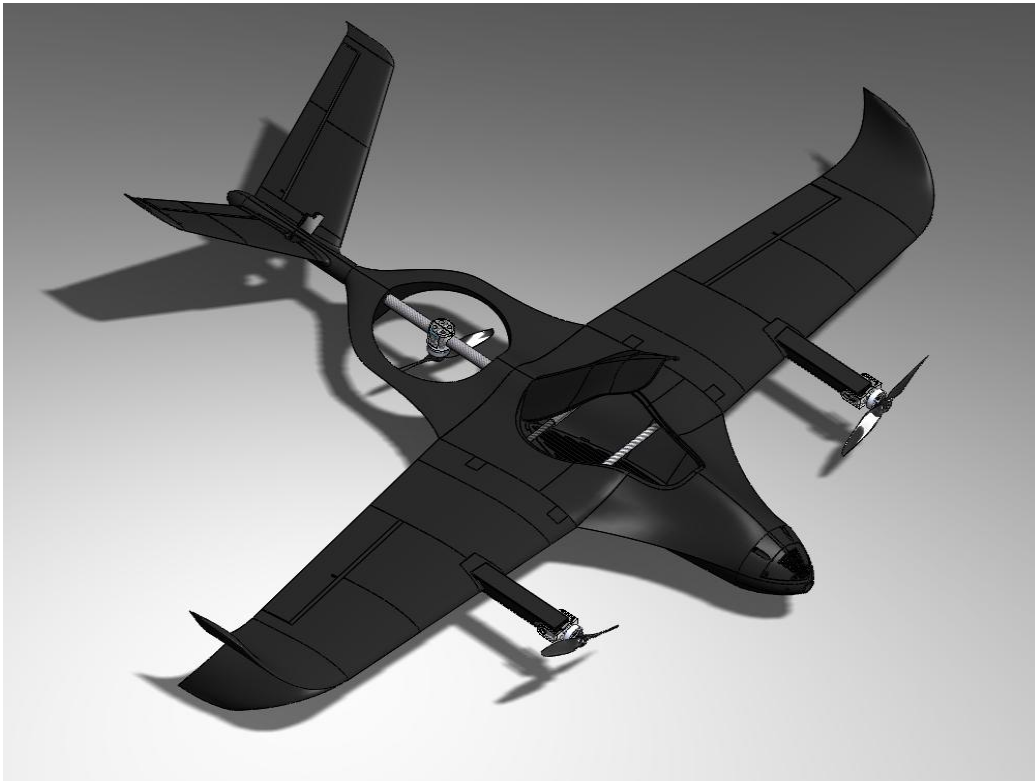


PROJECT ORCA

Fixed-Wing eVTOL Reconnaissance UAV



AEM 404 · Senior Capstone Design

The University of Alabama · Department of Aerospace Engineering & Mechanics

Spring 2026

Chief Engineer & CFD/Aerodynamics Lead: Fourie van Rooyen

Faculty Advisor: Dr. Jinwei Shen

Executive Summary

Project ORCA is a fixed-wing electric Vertical Takeoff and Landing (eVTOL) tiltrotor UAV developed as a University of Alabama AEM 404 Senior Capstone project. The aircraft is designed for reconnaissance, surveillance, and intelligence-gathering missions, combining vertical takeoff and landing capability with aerodynamically efficient fixed-wing cruise. Built on a Flightory Stallion V2 airframe platform, ORCA integrates a three-motor tiltrotor quadplane configuration running ArduPlane QuadPlane firmware.

The program advanced through a full NASA NPR 7123.1D-aligned design lifecycle: Mission Concept Review (MCR), Preliminary Design Review (PDR), Critical Design Review (CDR), Test Readiness Review (TRR), and Flight Readiness Review (FRR). Two flight-test campaigns were conducted in April 2026 at a UA-approved outdoor UAV test area. The aircraft achieved stable VTOL hover, completed one successful full transition cycle (VTOL → cruise → VTOL), and generated significant flight log data enabling in-depth post-flight analysis.

Key Results

- ✓ Stable VTOL hover demonstrated — 5–10 min total accumulated hover time
- ✓ First successful VTOL→cruise→VTOL transition cycle completed
- ✓ Full ANSYS CFD simulation at cruise (17 m/s): $L/D \approx 10.5\text{--}11.2$, $C_L = 0.58$ at $\alpha = 4^\circ$
- ✓ ANSYS FEA structural validation: $FoS \geq 2.0$ on primary structure
- ✓ ArduPlane QuadPlane firmware configured, tuned, and flight-validated
- ✓ FAA Part 107 compliant design throughout — $MTOW \leq 5$ kg

Team & Project Overview

Team Structure

Chief Engineer / CFD Lead	Fourie van Rooyen
Project Manager / GNC Lead	Matthew Cotting
Propulsion & Power / Safety Officer	Cohen McNabb
Structures Lead / CFO	Coralie Lallier
CAD Lead / Electronics Lead	Colin Thomasma
Technical Writer / Systems Lead	Eliel Smith
Pilot / Outreach Liaison	Keelan O'Doherty
Faculty Advisor	Dr. Jinwei Shen

Mission Objectives

The ORCA UAV was designed to accomplish the following primary objectives:

- Provide ≥ 30 minutes endurance at 17 m/s cruise, with ≥ 10 minutes hover capability
- Operate fully autonomously with waypoint navigation, VTOL transition, and return-to-home
- Carry modular payloads up to 2.0 kg including thermal/IR/EO imaging systems
- Launch and recover without a runway using vertical lift rotors
- Maintain full FAA Part 107 compliance (MTOW ≤ 55 lb, ≤ 400 ft AGL, LOS operation)

Vehicle Configuration Summary

Configuration	Fixed-wing with distributed VTOL lift rotors (tiltrotor tricopter)
Airframe Platform	Flightory Stallion V2
Wingspan	1.33 m
Length	~1.0 m
MTOW	≤ 5 kg
Cruise Speed	17 m/s (≈ 38 mph)
Target Endurance	60 minutes cruise
VTOL Hover Time	5–10 minutes at MTOW
Payload Capacity	≤ 2.0 kg modular EO/IR/night-vision
Propulsion	3× Thobby F60 tiltrotor motors + rear hover motor
Flight Controller	Matek F405-Wing V2 — ArduPlane QuadPlane firmware
Power System	6S LiPo (Zeee 6S 5200 mAh)

Construction	3D-printed ABS / LW-PLA + carbon fiber reinforcement
Regulatory Compliance	FAA Part 107

Aerodynamics & CFD Analysis

Airfoil Selection

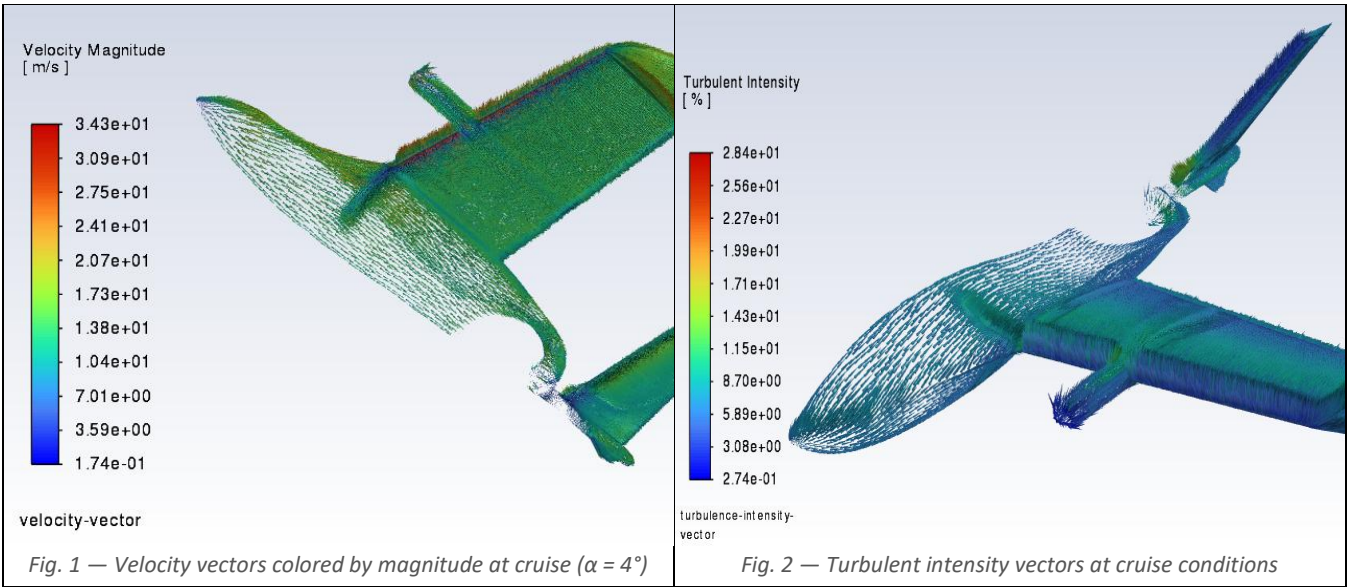
The ORCA wing uses the Eppler E205 airfoil, selected for its high lift-to-drag ratio at low Reynolds numbers ($Re \approx 2.3 \times 10^5$), predictable stall behavior, moderate pitching moment, and proven UAV performance record. The wing uses a high-aspect-ratio planform ($AR \approx 4.5$) to maximize endurance efficiency.

Aerodynamic Coefficients

Lift Curve Slope	$a \approx 0.088 / \text{deg} (\approx 5.0 \text{ rad}^{-1})$
Cruise $C_L (\alpha = 4^\circ)$	0.58
Zero-Lift Drag (C_{D0})	0.028
Oswald Efficiency (e)	0.78
Aspect Ratio (AR)	4.5
Predicted Cruise L/D	10.5–11.2
Stall Onset (α_{stall})	$\sim 10.5^\circ\text{--}11.2^\circ$
Cruise Margin to Stall	$>50\%$ (cruise $\alpha \approx 4^\circ$)

Full 3-D ANSYS Fluent CFD Simulation

A full 3-D CFD analysis was conducted in ANSYS Fluent 2025 R2 at cruise conditions ($V_\infty = 17 \text{ m/s}$) across angles of attack from 0° to 10° . The simulation used a viscous $k-\omega$ SST turbulence model with a fully resolved boundary layer mesh. Results confirm smooth, attached flow at cruise, strong upper-surface suction consistent with efficient lift generation, and no adverse pressure gradients at operating conditions.



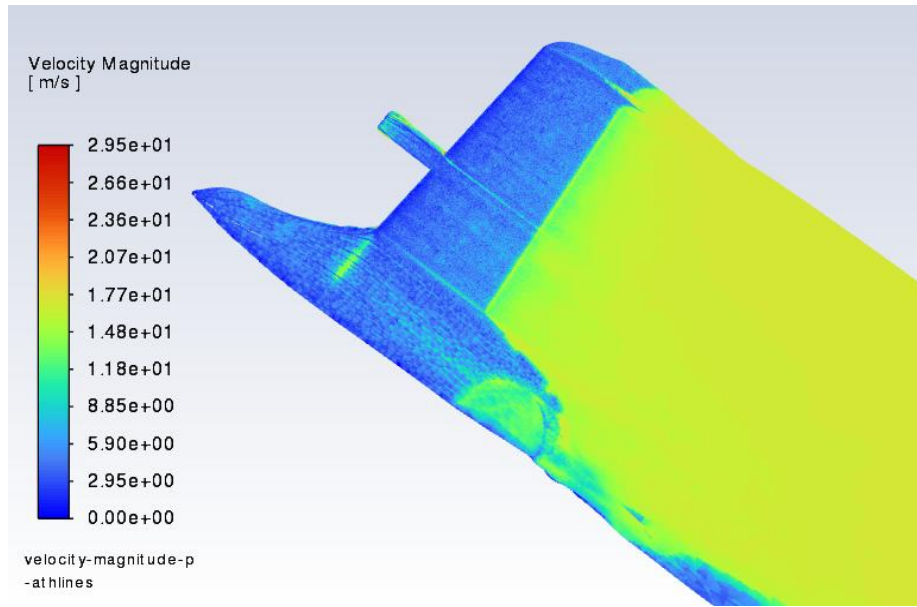


Fig. 3 — Velocity magnitude flow pathlines — isometric view showing acceleration over lifting surfaces

CFD Force/Moment Table at 17 m/s

AoA (°)	Lift (N)	Drag (N)	Moment (N·m)	C _L	C _D	C _M
0°	11.09	2.89	1.54	0.154	0.0401	0.0216
2°	29.59	0.48	8.44	0.394	0.0352	0.0597
4°	43.65	4.32	6.82	0.581	0.0575	0.0918
6°	47.77	8.24	7.10	0.636	0.1097	0.0956
8°	54.67	12.14	8.38	0.728	0.1617	0.1128
10°	51.62	16.44	7.89	0.688	0.2189	0.1061

Table 1 — Full-plane CFD simulation results at $V_\infty = 17$ m/s (ANSYS Fluent 2025 R2)

Structures & Finite Element Analysis

Structural Design Philosophy

The ORCA structural architecture uses a multi-material 3D-printed approach balancing weight, strength, and manufacturability. The design philosophy centers on lightweight thin-walled shells with internal ribs and spars, continuous load paths through all structural members, modular replaceability of all major assemblies, and composite reinforcement in high-load regions.

Material Selection

Primary Structure (motor mounts, brackets)	ABS — 30–35 MPa tensile, 1.8–2.2 GPa modulus, Tg ~105°C
Airframe Skins, Wings, Fuselage	LW-PLA-HT — ultra-lightweight, ~25 MPa tensile, Tg 65–70°C
Wing Spars & Motor Arms	Carbon fiber tubes — high stiffness-to-weight, bending resistance
High-stress joints	Epoxy & fiber tape — delamination and crack prevention
Infill Strategy (most parts)	7% infill — crush-tested superior to 10% (29.7 lbf vs. 17.8 lbf failure)
Motor mounts infill	≥ 25% — higher density required for thrust loads

ANSYS FEA Results

Structural finite element analysis was conducted in ANSYS Mechanical 2025 R2 covering three primary load cases: cruise aerodynamic loads, VTOL hover thrust loads, and gust loading at 15 kt.

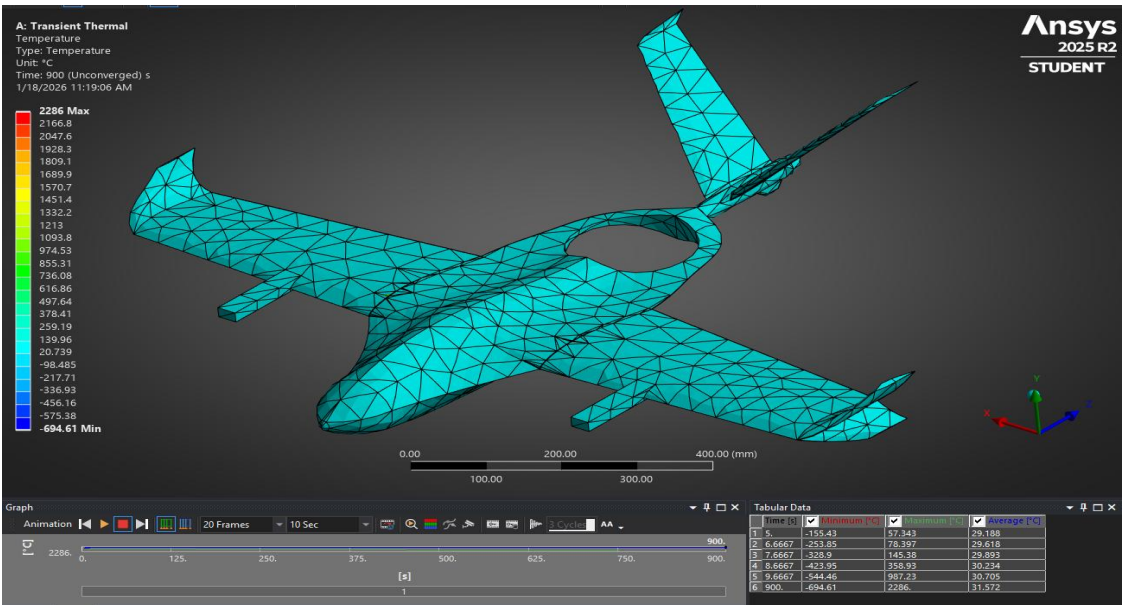


Fig. 4 — ANSYS Transient Thermal simulation of ORCA airframe (temperature distribution, 900 s exposure)

Wing max von Mises stress	< 18 MPa — FoS: 2.1–2.4
Wing tip deflection (15 kt gust)	4–6 mm — acceptable for UAV class
Fuselage stress at motor mounts	22–26 MPa — FoS ≥ 1.6
VTOL arm peak stress	~ 19 MPa — FoS ≈ 1.6 with carbon reinforcement
Global structural FoS	≥ 2.0 on primary structure (meets CDR requirement)
Landing/handling load peak	28–32 MPa — mitigated by reinforced belly skin

Stability, Control & GNC

Static Stability

The ORCA design satisfies required stability margins in all axes across three flight regimes: VTOL hover, transition, and fixed-wing cruise.

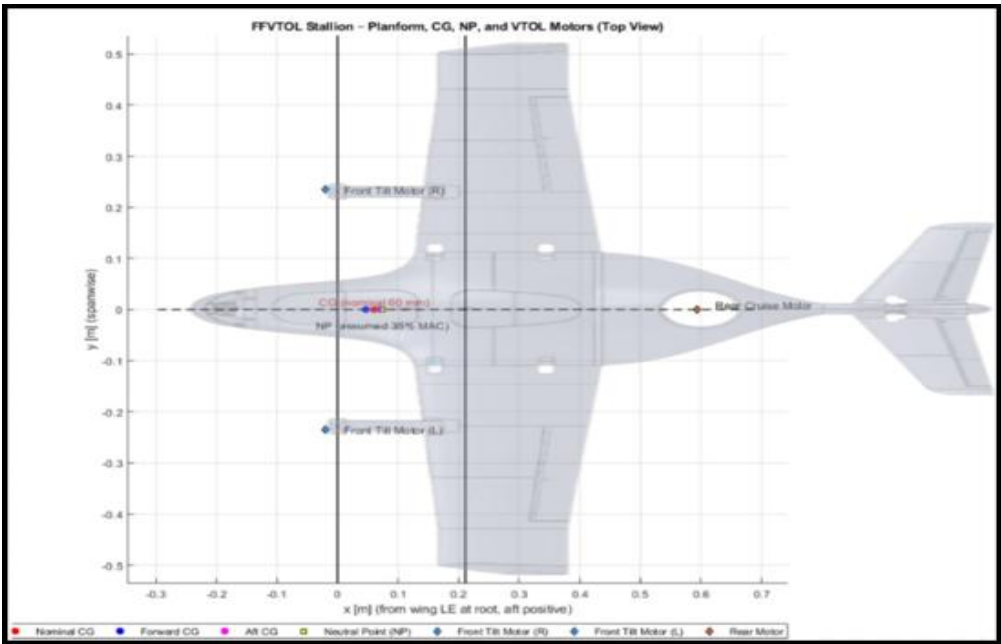


Fig. 5 — Static stability diagram: CG, NP, and VTOL motor positions relative to airframe planform

Longitudinal stability ($C_{m\alpha}$)	≈ -0.043 /deg — Stable ($C_{m\alpha} < 0$)
Static margin	8–10% MAC (target: 5–15%)
Directional stability ($C_{n\beta}$)	> 0 — Stable
Roll stability ($C_{l\beta}$)	< 0 — Wing-leveling
CG empty airframe	$\sim 23\%$ MAC
CG with full payload	$\sim 29\%$ MAC
VTOL CG alignment error	< 3 mm (acceptable)

Dynamic Stability Modes

Phugoid Mode	Period ~ 30 – 40 s, $\zeta \sim 0.05$ – 0.12 (lightly damped, normal for small UAVs)
Short Period Mode	Period ~ 0.3 – 0.6 s, $\zeta \sim 0.35$ – 0.45 (strongly stable)
Dutch Roll	$\zeta \approx 0.2$ – 0.3 — mitigated by yaw-damping autopilot gains

Roll Mode	Fast, strongly damped ($\zeta > 0.5$)
Spiral Mode	Slightly divergent — slow and manageable with autopilot

Transition Control Logic

VTOL-to-cruise transition uses a cubic blend scheduling function $\lambda(s) = 3s^2 - 2s^3$ where s is normalized transition time. This provides smooth thrust handoff, prevents step changes in angle of attack, and eliminates undesirable oscillations. Key ArduPlane QuadPlane parameters:

Q_TILT_MAX	45° — maximum tilt angle for transition
Q_TILT_RATE_UP	15°/s — motor tilt rate
Q_TRANSITION_MS	7000 ms — transition smoothing duration
AIRSPEED_MIN	12 m/s — minimum speed for fixed-wing transition gate
AIRSPEED_MAX	25 m/s — cruise speed limit
Q_M_BAT_VOLT_MAX/MIN	16.8 V / 13.2 V — battery voltage compensation

GNC Architecture

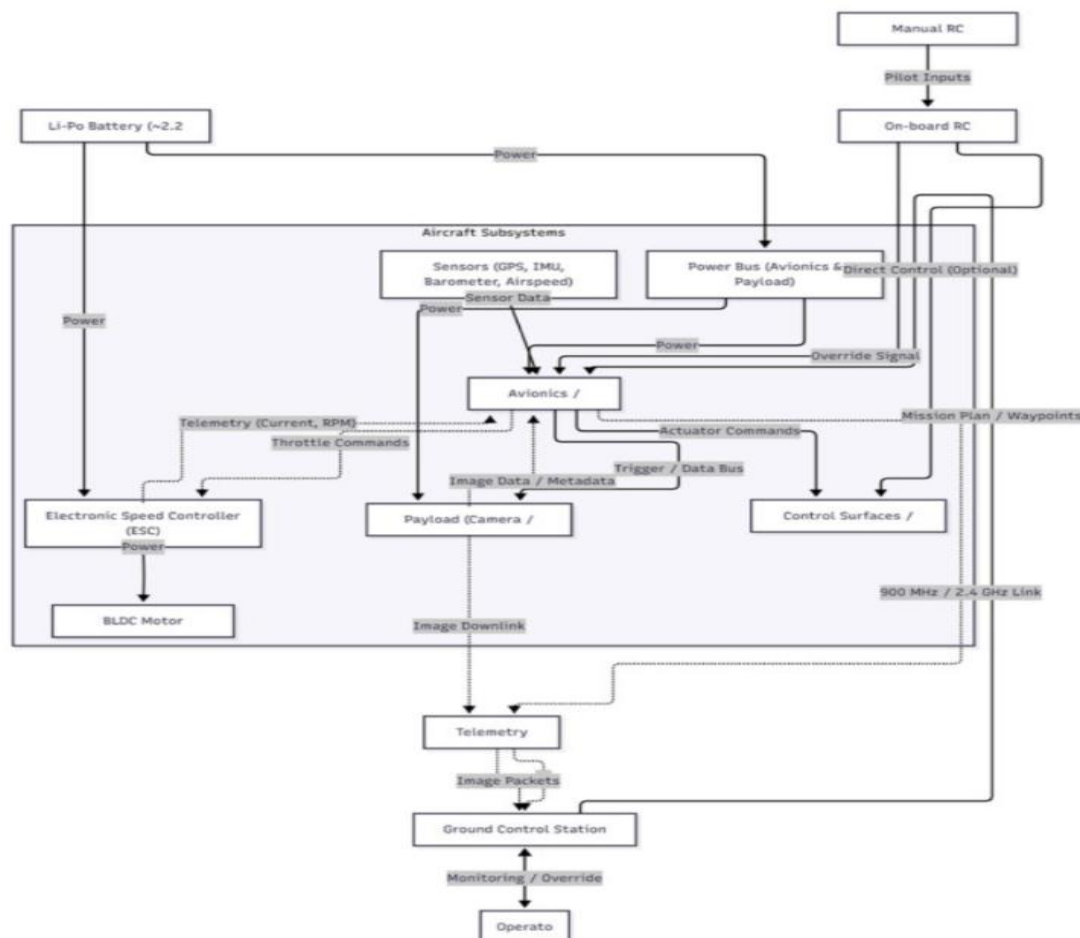


Fig. 6 — ORCA avionics and GNC system block diagram

Propulsion System

Architecture

ORCA uses a distributed electric propulsion system with three Thobby F60 tiltrotor motors configured as a tricopter quadplane. Two front motors tilt from vertical (VTOL) to horizontal (cruise) via servo-actuated mounts, while a fixed rear motor provides supplemental hover thrust and is disabled in cruise. All motors are driven by BLHeli32 ESCs over DSHOT600 protocol.

VTOL Motors	3× Thobby F60 — 1750 KV class, matched to 7×4.5" props
ESC Protocol	DSHOT600 — BLHeli32
VTOL T/W Ratio	1.41 at MTOW (5 kg) — exceeds EDS minimum of 1.3
Hover Power (estimated)	1,650–2,300 W total
Cruise Motor	7" high-efficiency propeller — rear-mounted
Cruise Power Draw	180–220 W at 17 m/s
Cruise Propeller Efficiency	$\eta_{prop} \approx 0.75$
Battery	Zeee 6S 5200 mAh LiPo
Estimated Cruise Endurance	35–65 min (analysis basis)
Tilt Servo Controller	PCA9685 I ² C expander (TIM8 conflict workaround)

Wiring & Power Distribution

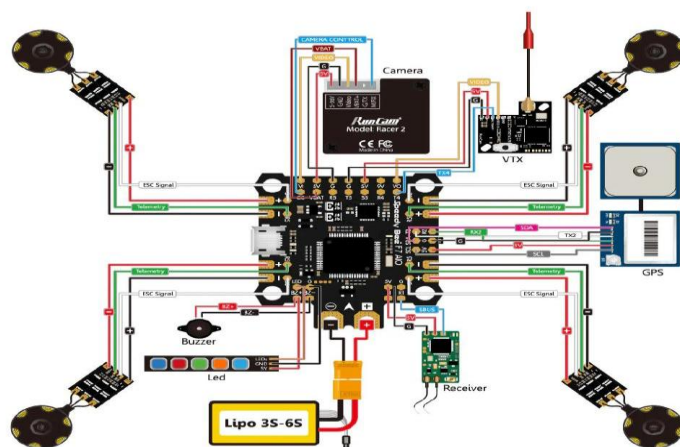


Fig. 7 — ORCA electronics wiring diagram — flight controller, ESCs, GPS, receiver, and payload

Physical Build & Manufacturing

Airframe Assembly

The ORCA airframe was manufactured entirely using FDM 3D printing on Bambu P1S and university Cube printers. The fuselage was printed in 3–4 modular segments and bonded using ABS-compatible epoxy with alignment pins. Wing panels use 7% gyroid infill LW-PLA-HT skins with ABS ribs and carbon fiber primary spars.

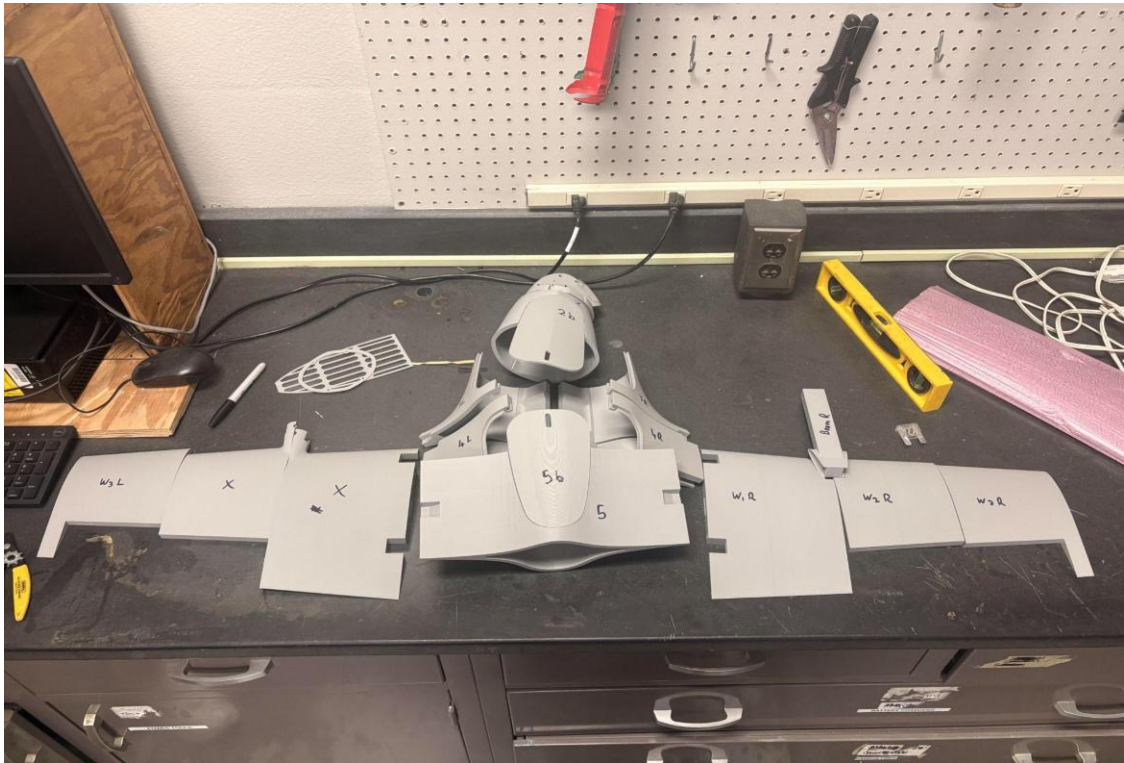


Fig. 8 — ORCA airframe laid out for pre-flight inspection — fuselage, wings, and VTOL boom assemblies

Manufacturing Parameters

Wing panels (LW-PLA-HT)	7% gyroid infill, nozzle 230–240°C, bed 50–60°C
Motor mounts (ABS)	25% infill, nozzle 240–250°C, bed 95–110°C
Fuselage (LW-PLA-HT)	3–4 segments, 7% infill, carbon longeron inserts
VTOL arms (LW-PLA-HT)	Carbon rod embedded during print pause
Total print time estimate	~60–65 hours across all components
Total material (ABS)	~400–600 g
Total material (LW-PLA)	~1,500–2,000 g

Flight Test Campaign & Results

Campaign Overview

Two flight-test sessions were conducted at a UA-approved outdoor UAV test area in April 2026. The campaign followed an incremental five-phase test plan (bench testing, integrated ground tests, VTOL hover, fixed-wing cruise, transition). The campaign is characterized as two distinct campaigns separated by a mid-program rebuild.

Session	Regime	Outcome	Key Observation
1A	VTOL Hover (QLOITER)	Crash (controlled)	FC commanded climb to match GPS home-altitude reference; kill switch at ~2.5 m AGL
—	Rebuild	—	Replace 2× front servos, 2× front motor mounts, rear motor, rear motor mount; epoxy fuselage and wingtips
2A	VTOL Hover (QSTABILIZE)	Met	Stable hover — 5–10 min total; less overcorrection than Session 1A
2B	Transition Cycle 1	Met ✓	VTOL→cruise→VTOL; aircraft swooped low during forward transition, recovered and returned
2C	Transition Attempt 2	Terminal crash	Lateral gust during tilt; UV-softened PLA booms; control overcorrection; terminal impact

Table 2 — High-level flight-test log summary (April 2026)

Session 2B: First Successful Transition

Session 2B represents the program's primary flight achievement. The aircraft was initiated from stable QSTABILIZE hover, transitioned to FBWA, executed the tilt sequence to 45°, accelerated to cruise airspeed, and then reversed the sequence to return to hover and land. The full VTOL→cruise→VTOL cycle was completed, validating the transition control logic and QuadPlane firmware configuration. Partway through transition, the aircraft pitched nose-down and descended, but recovered in response to aft stick input and completed its forward pass.

Root Cause Analysis — Session 2C Terminal Crash

Primary Failure Factors

- (1) UV-induced thermal softening of PLA boom arms — glass transition $T_g \sim 60^\circ\text{C}$ exceeded in direct sunlight
- (2) Compliant structure reduced GNC corrective authority — control law demands more, structure flexes more, diverges
- (3) Default ArduPlane PID gains untuned to this airframe — gains assumed rigid-body dynamics, airframe was softened
- (4) Cumulative fatigue from Session 1A impact — reduced structural reserve on rebuilt airframe

Quantitative Performance vs. CDR Predictions

Hover T/W ratio	CDR: 1.41 — Observed: Confirmed (adequate hover at MTOW)
Hover stability (±3°)	CDR: Met — Session 2A: Met in calm wind
Transition completion	CDR: Planned — Session 2B: Achieved (first cycle)
Cruise endurance	CDR: 35–65 min — Status: Not verified (program terminated)
Structural failure	CDR risk accepted — Root cause: UV/heat PLA degradation

Lessons Learned & Recommendations

Design Recommendations

- Replace PLA boom arms with PETG or fiber-reinforced ASA — UV and heat resistance critical for outdoor operations
- Perform formal PID tuning on-airframe before transition testing — default ArduPlane gains insufficient for this configuration
- Establish EKF convergence verification procedure before committing to altitude-hold modes (QLOITER)
- Add GPS home-altitude pre-arm check and altitude offset correction to preflight checklist
- Structural FEA should model UV-degraded material properties for worst-case outdoor analysis

Operational Improvements

- Tethered hover testing prior to free flight — would have revealed PID gain issues at lower risk
- Separate hover validation day from transition testing day — avoid stacking risk in single test window
- Dedicate time to formal PID autotuning using ArduPilot's onboard autotune function
- Apply UV-resistant coating or use ASA for all sun-exposed structural components

What Was Validated

Confirmed Technical Achievements

- ✓ QuadPlane firmware configuration with tiltrotor tricopter geometry — operational
- ✓ TIM8 timer conflict resolved via PCA9685 I²C expander — DSHOT and PWM coexistence confirmed
- ✓ VTOL hover stability in QSTABILIZE — adequate for test program
- ✓ Transition control logic functional — successful VTOL↔cruise cycle demonstrated
- ✓ Full CDR design cycle completed — 3D CFD, FEA, stability analysis, EDS compliance
- ✓ FAA Part 107 operations maintained throughout — no third-party property or personnel affected

Appendix: Engineering Design Specifications

EDS Compliance Summary

Requirement	CDR Status	Verification Method
60 min cruise endurance	On Track	Analysis & flight test
5–10 min hover	On Track	Thrust/power testing
2 kg payload	On Track	Mass budget & structural analysis
VTOL capability (< 5×5 m area)	On Track	Power/RPM → hover model
FAA Part 107 compliance	Met	Documentation
Stability ±3°	On Track	GNC tuning
CG within 23–33% MAC	Met	CAD & mass properties
Manufacturing requirements	Met	Printability analysis
Structural FoS ≥ 2	Met	ANSYS FEA
T/W ≥ 1.3 (VTOL)	Met	Thrust stand testing (1.41 achieved)

This document was prepared as a portfolio artifact summarizing Project ORCA (AEM 404 Senior Capstone, University of Alabama, Spring 2026). It consolidates deliverables spanning the Mission Concept Review through the Post-Flight Analysis Report.